

Part: A

1 ☐ Explain the Doppler Effect in detail. Derive the general expression for the apparent frequency when:

- The source is moving and the observer is stationary
- The observer is moving and the source is stationary

☐ Doppler Effect:

The Doppler Effect refers to the apparent change in frequency or wavelength of a wave due to the relative motion between the source of the wave and the observer. It is commonly observed in sound waves, where the pitch of a sound appears higher as the source approaches the observer, and lower as it moves away.

Here,

f = actual frequency of the source

f' = apparent frequency

v = velocity of the wave in the medium

v_o = velocity of the observer

v_s = velocity of the source

☐ System 1 (Source moving, observer stationary)

If the source moves with velocity v_s , the effective wavelength becomes,

$$\lambda' = \frac{v \pm v_s}{f}$$

The apparent frequency is,

$$\begin{aligned} f' &= \frac{v}{\lambda'} \\ \Rightarrow f' &= \frac{v}{\frac{v \pm v_s}{f}} \\ \Rightarrow f' &= f \cdot \frac{v}{v \pm v_s} \end{aligned}$$

(+) $\Rightarrow$ source moves away from observer

(-) $\Rightarrow$ source moves towards observer

□ System 2 (source stationary, observer moving)

The relative velocity of the wave with respect to the observer is $v \pm v_o$. Thus,

$$\begin{aligned} f' &= \frac{v \pm v_o}{\lambda} \\ \Rightarrow f' &= \frac{v \pm v_o}{\frac{v}{f}} \quad \left[\because \lambda = \frac{v}{f} \right] \\ \Rightarrow f' &= f \cdot \frac{v \pm v_o}{v} \end{aligned}$$

(+) $\Rightarrow$ observer moves towards source

(-) $\Rightarrow$ observer moves away from source

Ans.

Part: B₂

2 □ Draw and explain wave front diagrams illustrating:

- Source approaching observer
- Source receding from observer

Based on the diagram, explain the variation in frequency

□ Source approaching observer:

The source moves towards the observer. Each new wave front is emitted from a position closer to the observer. Wavefronts become closer together (compressed)

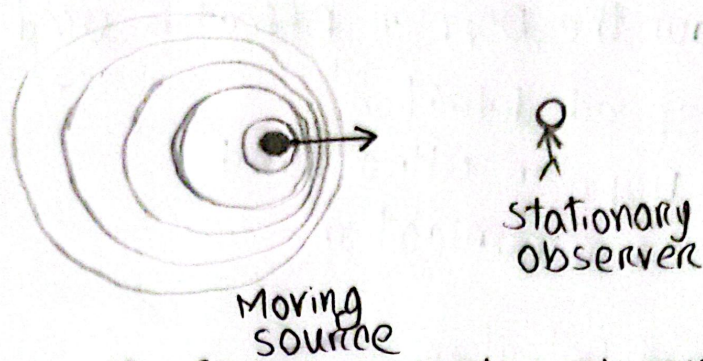


Fig: Source approaching observer

Here,

Wavelength decreases ($\lambda \downarrow$)

Since $f = \frac{v}{\lambda}$, frequency increases ($f \uparrow$)

- Observer hears a higher pitch

□ Source receding from observer:

The source moves away from observer. Each new wavefront is emitted farther from the observer. Wavefronts become spread out (expanded)

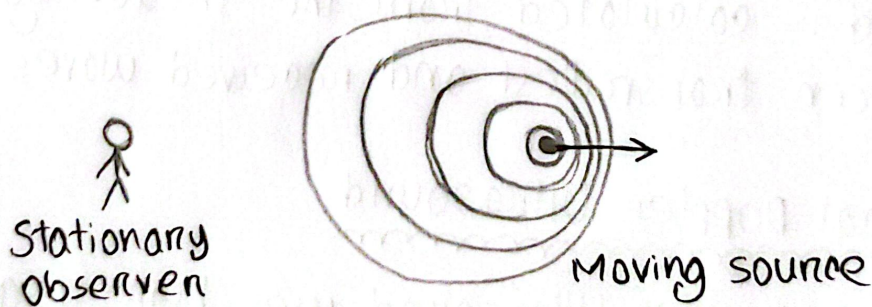


Fig: Source receding from observer

Here,

Wavelength increases ($\lambda \uparrow$)

Since, $f = \frac{v}{\lambda}$, frequency decreases ($f \downarrow$)

- Observer hears a lower pitch

Part: C

3 □ Explain how the Doppler Effect is used to determine:

- Radar speed detection
- Medical Doppler ultrasound
- Satellite communication

□ Radar Speed Detection:

Radar speed detection operates on the principle that electromagnetic waves reflected from a moving vehicle undergo a frequency shift due to the Doppler effect. The radar device emits waves toward the vehicle and the reflected waves are received back with a shifted frequency proportional to the vehicle's velocity.

- Approaching vehicle $\rightarrow$ frequency increases
- Receding vehicle $\rightarrow$ frequency decreases
- Speed is calculated from the frequency shift between transmitted and received waves

□ Medical Doppler Ultrasound:

Medical Doppler Ultrasound uses high-frequency sound waves that reflect from moving blood cells inside the body. The change in frequency of the reflected waves helps determine the velocity and direction of blood flow.

- Blood moving towards probe $\rightarrow$ higher frequency
- Blood moving away $\rightarrow$ lower frequency
- Used to detect blockages, measure heart function and monitor fetal circulation

■ Satellite Communication:

In satellite communication, relative motion between satellites and ground stations causes a Doppler shift in radio signals. This shift must be accounted for to ensure accurate transmission and reception of signals.

- Satellite approaching $\rightarrow$ frequency increases
- Satellite moving away $\rightarrow$ frequency decreases
- Used in GPS systems and signal correction for stable communication